A Dynamic General Equilibrium Model of Roads

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1.1 Representative Household

The economy is populated by a representative, infinitely–lived household that maximizes utility over consumption[1], housing services, and local amenities while disliking labor (including commuting time). In its most general form, the household's problem is to choose sequences of consumption $\{c_t\}_{t=0}^{\infty}$, housing stock $\{h_{t+1}\}_{t=0}^{\infty}$, private capital $\{K_{t+1}\}_{t=0}^{\infty}$, and labor supply $\{n_t\}_{t=0}^{\infty}$ to maximize lifetime utility:

$$\max_{\{c_t,h_t,n_t,K_{t+1},h_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \, \tilde{u}\big(c_t, h_t, n_t, G_t\big), \qquad 0 < \beta < 1, \tag{1}$$

We consider the following utility function,

$$\tilde{u}\big(c_t, h_t, n_t, G_t\big) \;=\; c_t \, h_t^{\alpha_h A(G_t)} \; \exp\!\Big[-\frac{\kappa}{1+\theta} \big(n_t \, [1+\phi(G_t)]\big)^{1+\theta}\Big] \tag{2}$$

where c_t denotes non–housing consumption, h_t the flow of housing services, and $A(G_t)$ the amenity derived from the public capital stock G_t, which represents the quality of local public goods.

After taking logs, we can write the utility function as a convenient separable specification

$$u\big(c_t, h_t, n_t, G_t\big) = \ln c_t + \alpha_h A(G_t) \ln h_t - \frac{\kappa}{1+\theta} \Big[n_t\big(1+\phi(G_t)\big)\Big]^{1+\theta}, \tag{3}$$

with n_t hours of market work, $\phi(G_t) \geq 0$ a commuting factor that rises as road quality falls, $\kappa > 0$, and $\theta > 0$ is the inverse Frisch elasticity of labor supply i.e. $\frac{1}{\theta}$ is the Frisch elasticity of labor supply with respect to real wages. The parameter α_h captures the household's preference for housing services, and $A(G_t)$ is a function that captures how public infrastructure quality (e.g., roads) affects the utility derived from housing. The term $\phi(G_t)$ captures the disutility of commuting, which increases as road quality deteriorates.

Budget Constraint and Capital/Housing Accumulation:

Let w_t be the real wage, r_t the rental rate on private capital K_t, $p_{h,t}$ the *pre–tax* purchase price of one unit of housing stock, τ_L the labor–income tax rate, and τ_h the property-tax rate levied on the *value of the housing stock*. The household's per-period budget can be written

[1] Since we don't have any random variables or distributions, we are not dealing with any expectations or expected utility.

in the following way:

$$c_t \ + \ i_t \ + \ p_{h,t}\, x_t \ = \ (1-\tau_L) w_t n_t \ + \ r_t K_t \ - \ \underbrace{\tau_h\, p_{h,t}\, h_t}_{\text{property tax on holdings}} \tag{4}$$

where x_t is net housing investment and i_t is private–capital investment. The capital and housing asset laws of motion are

$$K_{t+1} = (1-\delta_K) K_t + i_t, \qquad \delta_K \in (0,1), \tag{5}$$

$$h_{t+1} = (1-\delta_h) h_t + x_t, \qquad \delta_h \in (0,1). \tag{6}$$

Equivalently, we can substitute out i_t and x_t to rewrite the budget constraint as,

$$c_t + K_{t+1} - (1-\delta_K)K_t + p_{h,t}[h_{t+1} - (1-\delta_h)h_t] = (1-\tau_L)w_t n_t + r_t K_t - \tau_h p_{h,t} h_t \tag{7}$$

Optimality Conditions:

Maximization of (3) subject to (4)–(7) yields:

1. **Consumption–Saving Euler Equation:** The household equalizes the marginal utility of consuming an additional unit today to the discounted utility of saving it until tomorrow. This yields:

$$U_c(c_t, h_t, A(G_t)) = \beta U_c(c_{t+1}, h_{t+1}, A(G_{t+1}))\big[(1-\tau_L) w_{t+1} \frac{\partial n_{t+1}}{\partial K_{t+1}} + r_{t+1} + 1 - \delta_K\big]. \tag{8}$$

which simplifies (under certainty and taking labor choices as separate) to

$$U_c(c_t, \cdot) = \beta U_c(c_{t+1}, \cdot)\big(1 + r_{t+1} - \delta_K\big) \tag{9}$$

In other words, the marginal rate of intertemporal substitution equals the net discounted return on capital after depreciation. This is the standard Euler equation ensuring optimal capital accumulation.

Moreover, given that $U_c = \frac{1}{c_t}$ from our utility specification, we have

$$\frac{1}{c_t} = \beta\, \frac{1}{c_{t+1}} \big(1 + r_{t+1} - \delta_K\big) \tag{10}$$

2. **Labor–leisure trade–off:** The household chooses labor n_t such that the marginal disutility of working (including commuting) equals the after-tax marginal benefit of earning wages. Formally,

$$-U_n(c_t, h_t, A(G_t)) = (1-\tau_L)w_t U_c(c_t, h_t, A(G_t)) \tag{11}$$

where U_n is the partial derivative of utility with respect to labor (negative due to disutility) and U_c is the partial derivative with respect to consumption. Using our utility form, we get

$$U_n = -\kappa\big[n_t(1+\phi(G_t))\big]^{\theta}(1+\phi(G_t)), \quad U_c = \frac{1}{c_t} \tag{12}$$

Thus, the labor optimality condition 11 becomes:

$$\kappa\big[n_t(1+\phi(G_t))\big]^{\theta}(1+\phi(G_t)) = \frac{(1-\tau_L)w_t}{c_t} \tag{13}$$

$$\kappa\big(1+\phi(G_t)\big)^{1+\theta} n_t^{\theta} = \frac{(1-\tau_L)w_t}{c_t} \tag{14}$$

$$n_t = \left[\frac{(1-\tau_L)\,w_t}{\kappa\,c_t\,\big(1+\phi(G_t)\big)^{1+\theta}}\right]^{1/\theta} \tag{15}$$

Intuitively, the marginal rate of substitution between leisure (or time) and consumption equals the real after-tax wage. An increase in commuting factor $\phi(G_t)$ (worse roads) raises the left-hand side (effective disutility of labor), leading the household to supply less labor for a given wage. This aligns with the idea that poor road quality reduces effective labor supply as households require higher compensation to work the same hours because commuting erodes their usable time.

3. **Euler equation for Housing:** From our utility function, we have $u_{c,t} = 1/c_t$ and $u_{h,t} = \alpha_h A(G_t)/H_t$. The FOC with respect to H_{t+1} gives

$$p_{h,t}\,u_{c,t} \;=\; \beta\Big[u_{h,t+1} \;+\; u_{c,t+1}\,(1-\delta_h-\tau_h)\,p_{h,t+1}\Big] \tag{16}$$

or, in consumption units,

$$p_{h,t} \;=\; \beta\,\frac{c_t}{c_{t+1}}\Big[(1-\delta_h-\tau_h)\,p_{h,t+1}+\alpha_h\,A(G_{t+1})\,\frac{c_{t+1}}{H_{t+1}}\Big] \tag{17}$$

$$p_{h,t} \;=\; \beta\Big[(1-\delta_h-\tau_h)\,\frac{c_t}{c_{t+1}}\,p_{h,t+1}+\alpha_h\,A(G_{t+1}(\tau_h))\,\frac{c_t}{H_{t+1}}\Big] \tag{18}$$

The equation 18 encapsulates how the current price of housing $p_{h,t}$ reflects the discounted value of future benefits from owning housing and the benefits associated with holding it. The bracketed terms are, respectively, the *after–tax, after–depreciation resale value* next period and the *service dividend* (marginal value of one more unit of housing services) next period. The first term inside the brackets represents the discounted future resale value of the housing asset after accounting for depreciation and property taxes. The second term captures the flow of housing services (amenities) derived from owning the house, adjusted for the quality of public infrastructure $A(G_{t+1}(\tau_h))$ and scaled by consumption per unit of housing stock. Notice that the amenity value $A(G_{t+1}(\tau_h))$ depends on the future road quality, which in turn is influenced by the current property tax τ_h through its effect on maintenance spending, as explained in subsequent subsections. A cut in τ_h today leads to lower maintenance M_t, causing G_{t+1} to decline over time, which reduces $A(G_{t+1}(\tau_h))$ and thus lowers the service dividend from housing. This mechanism captures how changes in public infrastructure quality, driven by tax policy, affect housing prices through their impact on the flow of housing services.

1.2 Representative Firm

A competitive firm produces output Y_t with Cobb–Douglas technology

$$Y_t = K_t^{\alpha_k}\,(\mathcal{A}_t L_t)^{1-\alpha_k}, \qquad 0<\alpha_k<1, \tag{19}$$

enforcing factor–price equalization

$$w_t = (1-\alpha_k)\frac{Y_t}{L_t}, \qquad r_t = \alpha_k\frac{Y_t}{K_t}. \tag{20}$$

Profit Maximization. The firm operates in a competitive market for inputs and output, so it hires labor and rents capital until factor prices equal their marginal products. In each period, the firm solves

$$\max_{K_t, L_t} \{F(K_t, L_t) - w_t L_t - r_t K_t\}, \tag{21}$$

taking wage w_t and capital rental r_t as given. The first-order conditions are:

1. **Labor demand:**

$$w_t = (1-\alpha_k) K_t^{\alpha_k} (\mathcal{A}_t L_t)^{-\alpha_k} = (1-\alpha_k)\frac{Y_t}{L_t}. \tag{22}$$

   This is the usual result that the real wage equals the marginal product of labor (MP_L).

2. **Capital demand:**

$$r_t = \alpha_k K_t^{\alpha_k - 1} (\mathcal{A}_t L_t)^{1-\alpha_k} = \alpha_k \frac{Y_t}{K_t}. \tag{23}$$

   Thus, the rental rate on capital equals the marginal product of capital (MP_K).

Under constant returns and competitive markets, the firm earns zero economic profit in equilibrium (all output is paid out to factors). This means

$$Y_t = w_t L_t + r_t K_t \tag{24}$$

in equilibrium. Because poor local roads can indirectly reduce effective labor supplied (households may choose smaller L_t) or even the productivity of labor (if $\mathcal{A}_t$ were to depend on G_t), the equilibrium wage and output will adjust accordingly. In our model, we focus on the household-side mechanism; we do not explicitly insert G_t into $F(\cdot)$ so $\mathcal{A}_t$ is treated as exogenous, but one could imagine an extension where $\mathcal{A}_t = \mathcal{A}(G_t)$ similar to household amenities function, making public infrastructure a productive externality that boosts TFP. That would reinforce the mechanism: a decline in G_t would lower productivity and thus lower wages from the firm side as well. For now, wages change endogenously mainly due to labor supply shifts in the simple model.

1.3 Government and Public Infrastructure

Taxation and Spending. The government collects taxes and uses the revenue to finance road maintenance. Taxes have two components: an exogenous property tax τ_h (subject to voter approval) and an endogenous tax (such as τ_L on labor income, or potentially a lump-sum tax) that provides additional revenue. Denote by T_t^{exo} the exogenous tax revenue from the voted levy, and T_t^{endo} the endogenous tax revenue from other sources. For example, if τ_L is a labor tax, then $T_t^{\text{endo}} = \tau_L w_t n_t$, which varies with the economy and is therefore "endogenous". The property tax revenue is $T_t^{\text{exo}} = \tau_h p_{h,t} h_t$ each period, directly tied to the housing market outcome.

The government budget constraint is:

$$T_t^{\text{endo}} + T_t^{\text{exo}} = M_t, \tag{25}$$

assuming all tax revenue is spent on road maintenance M_t. We abstract from any other government consumption or transfers.

In a scenario where the road tax is cut (e.g., τ_h drops to zero after a failed levy vote), T_t^{exo} falls exogenously. For a balanced budget, maintenance spending M_t must be reduced by an equal amount. This captures the immediate fiscal impact of the tax cut, such as an observed 11% loss in road maintenance funding empirically.

Public Capital (Roads) Accumulation. The stock of public road infrastructure G_t evolves over time depending on maintenance. We adopt a law of motion in line with Rioja (2003) to formalize how insufficient maintenance leads to infrastructure depreciation, whereas sufficient maintenance can maintain road quality[2]. Let δ_G be the "normal" depreciation rate of roads (wear-and-tear if adequately maintained). Actual effective depreciation $\delta_{G,t}$ can exceed δ_G if maintenance is below the requirement. A convenient formulation is:

$$G_{t+1} = G_t(1 - \delta_{G,t}), \tag{26}$$

where

$$\delta_{G,t} = \varphi \cdot \max\left\{0, 1 - \frac{M_t}{\delta_G G_t^{\psi}}\right\}. \tag{27}$$

In words, if maintenance spending M_t falls short of the amount needed to cover normal depreciation ($\delta_G G_t$), the depreciation rate increases proportionally to the shortfall. The parameter $\varphi \geq 0$ governs how sensitive the infrastructure is to under-maintenance. For example, if M_t is only half of $\delta_G G_t$, then $1 - \frac{M_t}{\delta_G G_t} = 0.5$, and $\delta_{G,t} = 0.5\varphi$; this means roads wear out. If M_t is zero (no upkeep), depreciation could jump substantially (if φ is large, G_t may deteriorate very quickly). On the other hand, if M_t is at least $\delta_G G_t$ (maintenance fully covers yearly wear), then $\delta_{G,t} = 0$ (no deterioration). We also assume $\delta_{G,t}$ cannot go below 0 even if M_t exceeds $\delta_G G_t$ (any extra maintenance beyond replacing depreciation might marginally improve roads but with diminishing returns). This Riojas-style law of motion captures the intuition that lack of maintenance shortens the life of existing public capi-

[2]In this setup, roads can only be maintained, not improved.

tal. Maintenance spending effectively slows down depreciation, preserving the infrastructure stock; cutting maintenance causes G_t to decline faster over time. Importantly, a reduction in the road tax τ_h translates to lower M_t and thus a gradual decline in G_t over several periods. The deterioration in road quality will typically become noticeable after a few periods of under-maintenance, consistent with evidence that the negative effects on road conditions (and subsequently house prices) emerge with a lag. In our model, that lag is endogenously determined by the above law of motion – if M_t drops at $t = 0$, G_t will depreciate a bit faster each period, compounding into a significant drop in infrastructure quality after some years.

1.4 Housing Market

Housing is a tradable asset in *fixed* aggregate supply $\bar{h}$[3]. With a representative household, market clearing is

$$h_t \;=\; \bar{H} \quad \text{for all } t. \tag{28}$$

Given (28), the price $p_{h,t}$ is determined by the housing Euler (17) and the paths of $\{c_t, G_t\}$. In steady state, substituting $c_{t+1} = c_t = c$, $h_{t+1} = h_t = h = \bar{H}$, $G_{t+1} = G_t = G$, and $p_{h,t+1} = p_{h,t} = p_h$ into (17) yields the user–cost formula:

$$p_h \;=\; \frac{\beta}{1 - \beta(1 - \delta_h - \tau_h)} \alpha_h \, A(G) \, \frac{c}{h}. \tag{29}$$

Better roads i.e. higher $A(G)$ higher raise p_h. Larger h reduces marginal service value (log utility), lowering p_h ceteris paribus.

The steady-state housing price formula in equation (29) reveals how the discount factor β interacts with depreciation δ_h and property taxes τ_h to determine asset valuation. The term $\frac{\beta}{1-\beta(1-\delta_h-\tau_h)}$ represents the present value multiplier that converts the flow of housing services into an asset price.

To understand this multiplier, consider that $1-\delta_h-\tau_h$ represents the net retention rate of housing value after accounting for physical depreciation and tax obligations. The household effectively loses fraction $\delta_h + \tau_h$ of the housing asset's value each period through wear and tax payments. The remaining fraction $1 - \delta_h - \tau_h$ carries forward to the next period, where it generates both service flows and continuation value.

The economic intuition is clear: higher property taxes τ_h reduce the net retention rate, making housing a less attractive store of value and lowering equilibrium prices. A patient household (high β) values future service flows more highly, increasing the multiplier and

[3] This means that the total quantity of housing is constant over time, and any changes in demand will only affect prices. We can also allow extensions to the model that incorporate changes in housing supply.

thus housing prices. Conversely, rapid depreciation δ_h erodes the asset's continuation value, reducing prices through the same channel.

This formulation also shows why property tax cuts can have ambiguous effects on housing prices in our model. While lower τ_h directly increases the multiplier and boosts prices, it also reduces maintenance funding, leading to infrastructure decay that lowers the service flow $\alpha_h A(G)$. The net effect depends on the relative magnitudes of these opposing forces and their timing.

1.5 Competitive Equilibrium

We now define a recursive competitive equilibrium for this economy. Given an initial private capital stock K_0 and initial public capital (road quality) G_0, and a sequence of exogenous tax policy $\{\tau_{h,t}\}$ (with a drop in τ_h at the time of the tax cut shock), a competitive equilibrium is a sequence of allocations $\{c_t, h_{t+1}, n_t, K_{t+1}, M_t, G_t\}_{t\geq 0}$ and prices $\{w_t, r_t, p_{h,t}\}_{t\geq 0}$ such that:

1. **Household Optimization:** Given prices and taxes, the representative household chooses $\{c_t, h_{t+1}, n_t, K_{t+1}\}$ to maximize its utility subject to its budget and accumulation constraints. The household's choice satisfies the first-order optimality conditions described above (Consumption and housing Euler equations, labor supply condition, and housing demand condition, etc.), and the transversality condition on capital holding.

2. **Firm Optimization:** The representative firm chooses inputs $\{K_t, L_t\}$ each period to maximize profit. In equilibrium, $L_t = n_t$ (labor market clears) and $K_t = K_t$ held by the household (capital market clears), and the wage and rental rates satisfy $w_t = F_L(K_t, L_t)$ and $r_t = F_K(K_t, L_t)$ as given by the Cobb-Douglas marginal products. This ensures the firm's first-order conditions are met and profit is zero.

3. **Government Budget and Public Capital:** The government budget constraint holds each period: $T_t^{\text{endo}} + T_t^{\text{exo}} = M_t$, with $T_t^{\text{exo}} = \tau_{h,t} p_{h,t} h_t$ and $T_t^{\text{endo}} = \tau_L w_t n_t$. The road maintenance M_t feeds into the law of motion for G_t: given G_t at the start of period t, the next period's public capital G_{t+1} is determined by $G_{t+1} = G_t(1 - \delta_{G,t})$ with $\delta_{G,t}$ increasing if M_t is insufficient (as specified above). The government sets M_t according to the available tax revenue each period.

4. **Market Clearing:** All markets clear in equilibrium:

    - **Goods market:** Output is used for private consumption, private investment, and public maintenance. The resource constraint each period is:

$$Y_t = c_t + i_t + M_t. \tag{30}$$

Total output Y_t produced by the firm equals the sum of household consumption c_t, private investment i_t, and government demand for road maintenance M_t.

- **Labor market:** $L_t = n_t$. The labor supplied by the household equals labor demanded by the firm. The real wage adjusts such that this holds.
- **Capital market:** The firm's capital input equals the household's capital stock: K_t (used in production) $= K_t$ (owned by household). The rental rate r_t adjusts to clear this market.
- **Housing market:** $h_t = \bar{H}$. The total housing demanded by the household equals the fixed housing stock. The house price $p_{h,t}$ adjusts each period to clear the housing market.

5. **Consistency:** The expectations of the household are consistent with realized outcomes. In a deterministic steady-state scenario, this means the household correctly anticipates the path of prices and G_t. In a perfect-foresight or rational expectations equilibrium, the household foresees that a cut in τ_h implies lower future M_t and thus a gradually declining G_t, factoring this into its decisions. The sequence $\{p_{h,t}\}$ reflects the present value of housing services given the expected decline in amenities.

1.6 Solving the model

The steady-state equilibrium can be characterized as a system of 9 equations in 9 unknowns. We solve for the following endogenous variables: $(c, n, K, Y, w, r, q, M, G)$, where $q \equiv (1 + \tau_h)p_h$ is the effective housing price and housing quantity is fixed at $h = \bar{H}$.

1.6.1 System of Equilibrium Equations

The steady-state equilibrium is determined by the following 9 independent equations:

1. **Euler equation (steady-state):** From the intertemporal optimality condition:

$$1 = \beta(1 + r - \delta_K) \quad \Rightarrow \quad r = \delta_K + \beta^{-1} - 1 \tag{31}$$

This pins down the rental rate r directly from parameters.

2. **Firm capital pricing:** From profit maximization:

$$r = \alpha_k \frac{Y}{K} \tag{32}$$

This relates the output-capital ratio to the rental rate.

3. **Firm wage pricing:** With labor market clearing $L = n$:

$$w = (1 - \alpha_k) \frac{Y}{n} \tag{33}$$

This relates the wage to output and labor supply.

4. **Production technology:** The Cobb-Douglas production function:

$$Y = K^{\alpha_k} (A \cdot n)^{1-\alpha_k} \tag{34}$$

This ties together output, capital, and labor.

5. **Labor-leisure trade-off:** From household optimization:

$$\kappa [1 + \phi(G)]^{1+\theta} n^{\theta} = \frac{(1 - \tau_L) w}{c} \tag{35}$$

This links labor supply to wages, consumption, and road quality through commuting costs.

6. **Housing price determination:** From the housing Euler equation in steady state:

$$q = \frac{\beta}{1 - \beta(1 - \delta_h - \tau_h)} \cdot \alpha_h A(G) \frac{c}{\bar{H}} \tag{36}$$

This determines the housing price from the present value of housing services, incorporating the discount factor and net retention rate.

7. **Government budget constraint:** Balanced budget condition:

$$M = \tau_L w n + \tau_h p_h \bar{H} \tag{37}$$

This defines maintenance spending from tax revenues.

8. **Road quality steady-state:** From the law of motion $G_{t+1} = G_t(1 - \delta_{G,t})$:

$$G = G(1 - \delta_G) \quad \text{where} \quad \delta_G = \max\left\{0, \varphi\left(1 - \frac{M}{\delta_G^{\text{norm}} G^{\psi}}\right)\right\} \tag{38}$$

   In the minimal-upkeep steady state, this simplifies to $M = \delta_G^{\text{norm}} G^{\psi}$.

9. **Resource constraint:** Goods market clearing with capital accumulation:

$$c = Y - \delta_K K - M \tag{39}$$

   Note that $i = K - (1 - \delta_K)K = \delta_K K$ in steady state. This accounts for the allocation of output between consumption, investment, and maintenance.

1.7 Transitional Dynamics and Welfare Analysis

To analyze the transitional dynamics following a tax cut, we solve the model numerically two different ways: under perfect foresight[4] and partial adjustment[5]. The key steps are:

1. **Calibration:** Assign parameter values based on empirical estimates or literature benchmarks. For example, β, α_h, η, κ, θ, δ_K, δ_G, and φ are calibrated to match observed household behavior, infrastructure depreciation rates, and labor supply elasticities.

2. **Initial Steady State:** Compute the pre-tax-cut steady state by solving the equilibrium conditions under the initial τ_h. This provides baseline values for $\{M_t, G_t, p_{h,t}, c_t, n_t, Y_t\}$.

3. **Shock Implementation:** Introduce the tax cut by reducing τ_h at $t = 0$. Solve the model forward to trace the path of endogenous variables $\{M_t, G_t, p_{h,t}, c_t, n_t, Y_t\}$ as the economy transitions to the new steady state.

4. **Dynamic Path:** Use one of the methods stated above to compute the full transition path of $\{M_t, G_t, p_{h,t}, c_t, n_t, Y_t\}$. Households anticipate the future decline in G_t due to lower maintenance and adjust their consumption, labor supply, and housing demand accordingly. The path of G_t is updated each period using the law of motion in equation 26.

[4] Households are forward-looking and adjust their behavior immediately to changes in policy and economic conditions.

[5] Households adjust their behavior gradually in response to changes in policy and economic conditions.

5. **Welfare Analysis:** Compute the household's lifetime utility before and after the tax cut. Decompose the welfare change into short-run gains (higher disposable income) and long-run losses (lower amenities and higher commuting costs).

6. **Sensitivity Analysis:** Vary key parameters (e.g., φ, α_h, θ) to test the robustness of the results. Examine how the magnitude of the tax cut or alternative government policies affect outcomes.

The numerical solution involves iterating on the household's Euler equations, firm conditions, and government budget constraint while updating G_t using its law of motion. This allows us to simulate the gradual decline in road quality and its impact on housing prices, labor supply, and overall welfare.

1.8 Discussion of Policy Shock

A permanent cut in the voted levy lowers τ_h exogenously. In the short run, disposable income rises as tax burden decreases, but G_t is initially unchanged. This leads to a sharp increase in consumption but also a sharp decrease in maintenance budget. Over time, reduced maintenance spending accelerates depreciation in equation 26, lowering $A(G_t)$ and raising $\phi(G_t)$; this depresses labor supply, housing demand, and therefore equilibrium house prices, as demonstrated in Section 1.10.

1.9 Model Calibration

The model parameters are calibrated based on empirical evidence and standard values from the literature. Table 1 summarizes the calibrated values used in the analysis. These values are chosen to reflect realistic economic conditions. For example, $\beta = 0.96$ corresponds to a 4% annual discount rate, while $\delta_K = 0.05$ reflects typical depreciation rates for private capital. Other parameters such as labor disutility parameters κ and θ are calibrated to match observed labor supply elasticities, and φ captures the empirical sensitivity of road quality to maintenance shortfalls.

1.10 Model Results

The model is solved numerically using both a perfect foresight and a partial adjustment algorithm to simulate the economy's response to a permanent cut in the road maintenance tax τ_h. Below, we present the key results from partial adjustment simulations[6].

[6] We focus on partial adjustment results here as they are better equipped for illiquid housing markets and gradual behavioral responses. Perfect foresight results are qualitatively similar but exhibit sharper short-run

Table 1: Baseline calibration and sensitivity ranges

Parameter	Model block	Baseline	Sensitivity range	Key empirical source
β	Household intertemporal utility	0.96	0.94–0.99	Cooley (1995)
α_h	Utility weight on housing services	0.35	0.25–0.40	(U.S. Bureau of Labor Statistics, 2024)
η	Amenity elasticity	0.88		†
θ	Inverse Frisch elasticity labor supply	2	1.5–4.0	Chetty et al. (2011)
κ	Labor-disutility scale	2.9	2.9	Chatterjee, Gibson and Rioja (2017)
α_k	Private capital share	0.30	0.25–0.40	Chatterjee, Gibson and Rioja (2017)
δ_K	Private capital depreciation (annual)	0.06	0.04–0.08	Cooley (1995)
δ_h	Housing depreciation (annual)	0.05	0.04–0.08	
δ_G^{norm}	Road stock depreciation	0.25		†
φ	Under-maintenance sensitivity	0.64	0.25–0.75	Rioja (2003)
γ	Commuting disutility sensitivity	0.47		†
ψ	Maintenance requirement elasticity	0.47		†
τ_L	Labor income tax share for roads	0.35		†

Note: † Calibrated to match empirical moments

- A reduction in τ_h from 1% to 0% leads to an decrease in maintenance budget M_t and road quality G_t, which declines gradually over time due to the law of motion for public capital.

- The house price $p_{h,t}$ first increases and then falls significantly as the amenity value $A(G_t)$ declines, reflecting the reduced desirability of housing in areas with deteriorating roads

The figures 1 illustrates the dynamics of key variables over time following the tax cut. The model's transitional dynamics following a permanent reduction in the property tax from 1 mills to 0 mills.

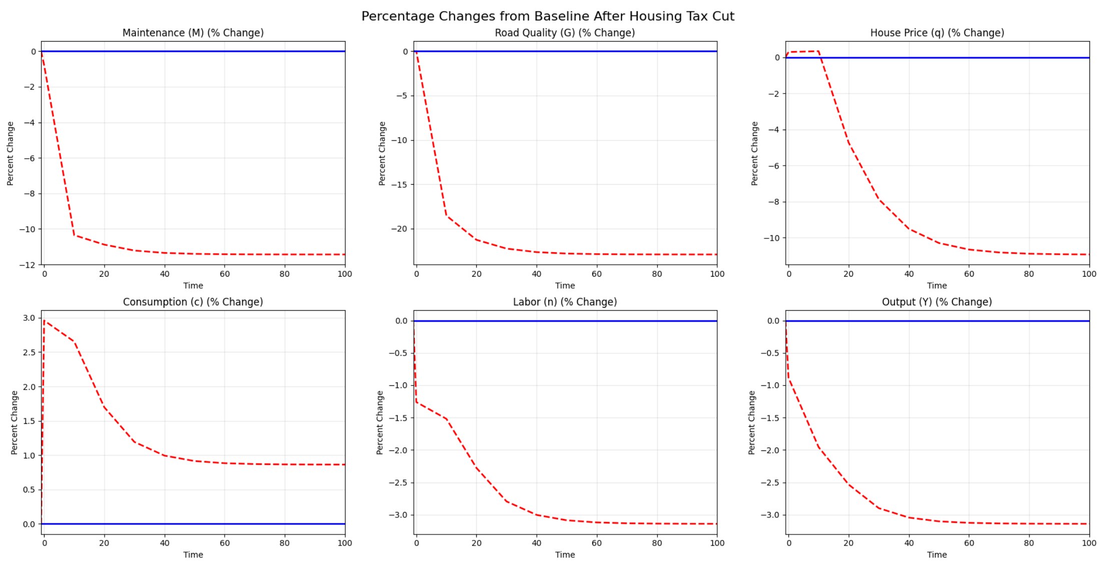

Figure 1: Dynamics of variables following the tax cut.

Figure 1 plots the full transition paths (percentage deviations from baseline). Most of the adjustment in M_t and q_t occurs early, while G_t converges more gradually as under-maintenance compounds over time. Findings from Figure 1 are summarized below:

- **Roads and maintenance.** Following the road tax cut, the road maintenance budget drops by about 10% and then drifts to roughly -11% (top-left panel of Figure 1). Given the law of motion for roads, G_t declines gradually and monotonically, approaching a long-run decline to 21% by the end of the simulated horizon.

- **House prices.** The housing price q_t initially shows a small blip (about +0.3%) and then declines smoothly toward a long-run drop of an average of 9% over the simulated horizon as the amenity value $A(G_t)$ deteriorates.

dynamics.

- **Aggregates.** The macro aggregates move modestly: labor n_t falls by about -3.0%, output also Y_t by about -3%, and consumption c_t increases by about 3% in the long run. Consumption exhibits a slight short-run overshoot (of +3%) as households initially respond to higher disposable income from the tax cut, before settling slightly above baseline (bottom row). This reflects the household's initial response to higher disposable income from the tax cut, followed by adjustments and substitution effect as road quality declines.

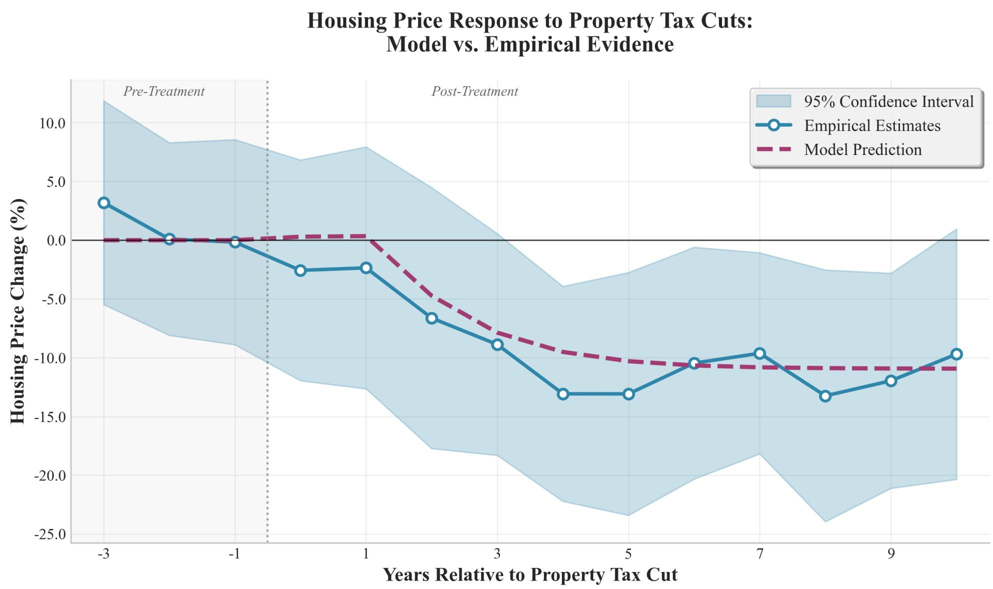

Figure 2: Model vs Empirical Estimates of House Prices Following the Tax Cut.

Model vs. empirical house-price path. Figure 2 compares the model-implied house-price response to the event-study estimates in Figure **??**. After aligning the time units, the model tracks the medium- and long-run decline well, squarely within the empirical confidence band. The model is slightly conservative initially as the drop in housing prices occurs more gradually than the empirical estimates. Nevertheless, the overall magnitude and timing of the decline remain consistent with the data as the model starts to deviate from the baseline around year 2 and approaches a long-run average drop of about 9% by year 10. This pattern is consistent with the mechanism presented in the paper: eliminating the tax reduces maintenance revenues, which gradually lowers road quality, depresses amenity $A(G_t)$, and in turn pulls down housing valuations.

Overall, the calibrated model reproduces the targeted steady-state effects (about -21% in roads, -9% in house prices, and -11% in maintenance) and generates transitional dynamics for house prices that closely follow the empirical path while remaining slightly less sharp at the short-run trough.

1.11 Model Limitations

1. **Representative agent assumption.** Heterogeneity in income or commuting preferences could attenuate price effects. Sensitivity analyses with quasi-linear preferences or idiosyncratic $\phi(G_t)$ show qualitative robustness.

2. **No spatial heterogeneity.** We currently ignore spatial heterogeneity. The model intentionally abstracts from space for tractability. Future work can embed G_t in a Rosen–Roback framework to capture cross-location sorting.

3. **Fixed housing stock.** Allowing new construction would introduce a supply elasticity that dampens price responses. Empirical evidence suggests housing supply in many U.S. jurisdictions is quite inelastic.

4. **No direct productivity role for roads.** Incorporating G_t into firm TFP $\mathbb{A}_t$ would further enrich the model. However, this would require additional assumptions about how road quality affects firm location and productivity, which is beyond our current scope.

The DGE framework formalizes the intuition that road–maintenance tax cuts provide immediate relief but, by undermining infrastructure quality, erode local amenities and raise commuting costs, culminating in lower house values and welfare.

1.12 Derivations of Key Equations

1.12.1 Derivation of the Euler Equation for consumption 10

Below $U_c \equiv \partial u/\partial c$ and λ_t is the Lagrange multiplier on the period-t budget constraint.

Set up the Lagrangian:

$$L = \sum_{t=0}^{\infty} \beta^t \Big[u(c_t, h_t, A(G_t)) + \lambda_t \Big((1-\tau_L) w_t n_t + r_t K_t - \tau_h p_{h,t} h_t - c_t - p_{h,t} h_t - K_{t+1} + (1-\delta_K) K_t \Big) \Big].$$

FOC w.r.t. consumption c_t:

$$\frac{\partial L}{\partial c_t} = 0 \quad \Longrightarrow \quad \beta^t U_c(c_t, h_t, A(G_t)) - \lambda_t = 0 \quad \Longrightarrow \quad \lambda_t = \beta^t U_c(c_t, h_t, A(G_t)). \qquad \text{(A1)}$$

FOC w.r.t. next-period capital K_{t+1}:

$$\frac{\partial L}{\partial K_{t+1}} = 0 \quad \Longrightarrow \quad -\lambda_t + \lambda_{t+1}\Big[(1-\tau_L)w_{t+1}\frac{\partial n_{t+1}}{\partial K_{t+1}} + r_{t+1} + 1 - \delta_K\Big] = 0. \tag{A2}$$

Combine (A1) and (A2):

Insert λ_t and λ_{t+1} from (A1) into (A2):

$$\beta^t U_c(c_t, h_t, A(G_t)) = \beta^{t+1} U_c(c_{t+1}, h_{t+1}, A(G_{t+1}))\Big[(1-\tau_L)w_{t+1}\frac{\partial n_{t+1}}{\partial K_{t+1}} + r_{t+1} + 1 - \delta_K\Big].$$

Divide both sides by β^t to obtain:

$$U_c(c_t, h_t, A(G_t)) = \beta U_c(c_{t+1}, h_{t+1}, A(G_{t+1}))\Big[(1-\tau_L)w_{t+1}\frac{\partial n_{t+1}}{\partial K_{t+1}} + r_{t+1} + 1 - \delta_K\Big].$$

Why the derivative $\frac{\partial n_{t+1}}{\partial K_{t+1}}$? Capital chosen today affects tomorrow's labor choice only through the household's optimal plan. If, as we assume, labor in each period is chosen independently of the inherited capital stock, this derivative is zero and the term drops out, giving the familiar

$$U_c(c_t, \cdot) = \beta U_c(c_{t+1}, \cdot)\big(1 + r_{t+1} - \delta_K\big),$$

which is the standard Euler equation for intertemporal consumption smoothing.

Equation 10 still contains the marginal utility of consumption $U_c(\cdot)$. Because the one-period utility function chosen is:

$$u(c_t, h_t, A(G_t)) = \ln c_t + \alpha_h \ln h_t + \eta \ln A(G_t) - \frac{\kappa}{1+\theta}\big[n_t(1+\phi(G_t))\big]^{1+\theta}, \tag{6}$$

its partial derivative with respect to consumption is:

$$U_c(c_t, h_t, A(G_t)) = \frac{\partial u}{\partial c_t} = \frac{1}{c_t} \quad \text{(log utility)}$$

Substituting U_c into the Euler Equation:

Start from Eq. 10:

$$U_c(c_t, \cdot) = \beta\, U_c(c_{t+1}, \cdot)\big(1 + r_{t+1} - \delta_K\big). \tag{10}$$

Replace U_c with $\frac{1}{c}$ (log utility):

$$\frac{1}{c_t} = \beta \frac{1}{c_{t+1}} \big(1 + r_{t+1} - \delta_K\big)$$

This is exactly Eq. 11:

$$\boxed{\frac{1}{c_t} = \beta \frac{1}{c_{t+1}} \big(1 + r_{t+1} - \delta_K\big)} \tag{11}$$

1.12.2 Labor–leisure trade–off

To derive the labor-leisure trade-off condition, we start by computing the partial derivatives of the utility function with respect to consumption and labor:

For consumption:

$$u_c(c_t, h_t, n_t, G_t) = \frac{1}{c_t} \tag{B1}$$

For labor, we have:

$$u_n(c_t, h_t, n_t, G_t) = -\,\kappa \big[n_t\big(1 + \phi(G_t)\big)\big]^{\theta} \big(1 + \phi(G_t)\big). \tag{B2}$$

Optimal labour supply equates the marginal disutility of work to the after-tax marginal benefit of wages:

$$-\,u_n(c_t, h_t, n_t, G_t) \;=\; (1 - \tau_L)\, w_t\, u_c(c_t, h_t, n_t, G_t)$$

Using B1 and B2, we can rewrite this as:

$$\kappa \big[n_t\big(1 + \phi(G_t)\big)\big]^{\theta} \big(1 + \phi(G_t)\big) \;=\; \frac{(1 - \tau_L)\, w_t}{c_t}$$

This is exactly the condition we wanted to derive:

$$\kappa \big(1 + \phi(G_t)\big)^{1+\theta} n_t^{\theta} \;=\; \frac{(1 - \tau_L)\, w_t}{c_t}$$

$$n_t \;=\; \left[\frac{(1 - \tau_L)\, w_t}{\kappa\, c_t\, \big(1 + \phi(G_t)\big)^{1+\theta}}\right]^{1/\theta}$$

1.12.3 Euler equation for Housing

We derive the housing Euler starting from the household's problem.

1. Primitives. Per–period utility:

$$u(c_t, h_t, n_t, G_t) = \ln c_t + \alpha_h A(G_t) \ln h_t - \frac{\kappa}{1+\theta}\big[n_t(1+\phi(G_t))\big]^{1+\theta}.$$

Marginal utilities (holding G_t fixed):

$$u_{c,t} = \frac{1}{c_t}, \qquad u_{h,t} = \frac{\partial u}{\partial h_t} = \alpha_h \frac{A(G_t)}{h_t}.$$

2. Budget constraint in stock form. Using $H_{t+1} = (1-\delta_h)H_t + x_t$ and substituting $x_t = H_{t+1} - (1-\delta_h)H_t$, the period-$t$ budget is

$$c_t + i_t + p_{h,t}\big(H_{t+1} - (1-\delta_h)H_t\big) = (1-\tau_L)w_t n_t + r_t K_t - \tau_h p_{h,t} H_t.$$

Rearrange the housing terms:

$$-p_{h,t}H_{t+1} + p_{h,t}(1-\delta_h)H_t - \tau_h p_{h,t} H_t = -p_{h,t}H_{t+1} + p_{h,t}(1-\delta_h-\tau_h)H_t.$$

3. Lagrangian. Let λ_t be the multiplier on the budget constraint:

$$\begin{aligned}\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ & u(c_t, h_t, n_t, G_t) \\ & + \lambda_t\big[(1-\tau_L)w_t n_t + r_t K_t - c_t - i_t - p_{h,t}(H_{t+1} - (1-\delta_h)H_t) \\ & \quad - \tau_h p_{h,t} H_t - (K_{t+1} - (1-\delta_K)K_t)\big]\Big\} \end{aligned} \tag{40}$$

4. FOCs for c_t and H_{t+1}. Consumption:

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 \;\Rightarrow\; \lambda_t = u_{c,t}.$$

For H_{t+1} (differentiating with respect to the housing stock chosen for next period):

$$\frac{\partial \mathcal{L}}{\partial H_{t+1}} = 0: \quad \beta^{t+1} u_{h,t+1} + \beta^{t+1}\lambda_{t+1} p_{h,t+1}(1-\delta_h-\tau_h) - \beta^t \lambda_t p_{h,t} = 0.$$

Divide by β^t and rearrange:

$$p_{h,t}\lambda_t = \beta\big[u_{h,t+1} + \lambda_{t+1} p_{h,t+1}(1-\delta_h-\tau_h)\big].$$

5. Replace multipliers with marginal utility of consumption. Since $\lambda_t = u_{c,t}$:

$$p_{h,t}u_{c,t} = \beta\Big[u_{h,t+1} + u_{c,t+1}(1-\delta_h-\tau_h)p_{h,t+1}\Big],$$

which is equation (16).

6. Convert to consumption units. Divide both sides by $u_{c,t+1}$:

$$p_{h,t}\frac{u_{c,t}}{u_{c,t+1}} = \beta\Big[(1-\delta_h-\tau_h)p_{h,t+1} + \frac{u_{h,t+1}}{u_{c,t+1}}\Big]$$

With $u_{c,t} = 1/c_t$ we have $\dfrac{u_{c,t}}{u_{c,t+1}} = \dfrac{c_{t+1}}{c_t}$, so

$$p_{h,t} = \beta\,\frac{c_t}{c_{t+1}}\Big[(1-\delta_h-\tau_h)p_{h,t+1} + \frac{u_{h,t+1}}{u_{c,t+1}}\Big].$$

Substitute $u_{h,t+1}/u_{c,t+1} = \big(\alpha_h A(G_{t+1})/H_{t+1}\big)\cdot c_{t+1}$:

$$p_{h,t} = \beta\,\frac{c_t}{c_{t+1}}\Big[(1-\delta_h-\tau_h)p_{h,t+1} + \alpha_h A(G_{t+1})\frac{c_{t+1}}{H_{t+1}}\Big]$$

which is equation (17).

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